# Experiment # 5

**"To Determine the Resistance of a Galvanometer by Half-Deflection Method".**

**Theory:** In the arrangement shown in Fig.1 if the shunt resistance S is very small compared to the galvanometer resistance G, then the potential difference (V) between the ends of the shunt resistance S remains nearly constant for all values of $R_1$.
Thus when R1=0, then the galvanometer current Cg is given

$$\frac{V}{G} = kd \text{.....(1)}$$

Where d is the deflection of the spot of light on the scale and k is the galvanometer constant. If now a resistance $R_1$ is introduced in the galvanometer circuit such that the deflection reduces to d/2, then

$$C'_g = \frac{V}{G+R_1} = k\frac{d}{2} \text{.....(2)}$$

Where $C'_g$ is the new galvanometer current in the changed circumstances. Dividing (1) by (2), we get

$$\frac{G+R_1}{G} = 2, or G + R_1 = 2G$$

or

$$G = R_1 \text{.....(3)}$$

Hence by simply measuring R1, G can be found out.

**Apparatus:**
Suspended coil galvanometer, shunt box S, resistance R and $R_1$, commutator K, cell E and connecting wires.

**Procedure:**
(i) Make connection as shown in Fig. 1. Bring one sharp edge of the spot of light at the zero mark of the scale.
(ii) Insert a resistance (R) of the order of 1000 ohms in the battery circuit. Make $R_1$=0 by putting all the plugs in the box. Beginning with the smallest value (S=0.1 ohm) of the

shunt resistance S, go on increasing S until you obtain a defection of about 10 cm on the scale. Note this deflection.

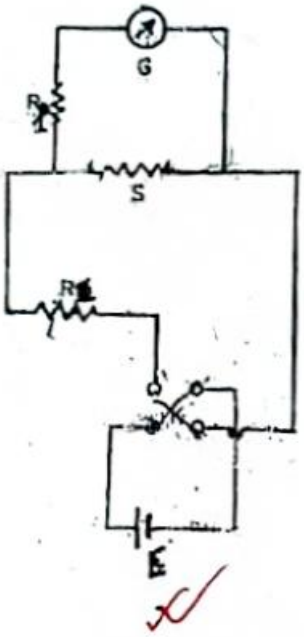


Fig. 1

(iii) Keeping the resistance R constant, adjust the value of the resistance $R_1$ until the deflection is reduced to half of the former. Record this value of $R_1$ which is the value of the galvanometer resistance G.

(iv) Stop the current in the circuit and examine if the same sharp edge of the spot of light is still at zero of the scale. If not, adjust the scale to bring it to zero. Make the value of $R_1$ zero and keep R the same. Now reverse the current with the commutator K. Repeat the whole operation to get another value of G.

(v) Keeping the value of the rsistance R the same, change the value of the shunt resistance S to obtain a different deflection of round about 10 cm and similarly determine the value of G.

(vi) Repeat the operation three times with different value of R in the battery circuit and two values of S for each R.

**Results:** In the following table R is the resistance in the battery circuit and $R_1$ is the resistance in the galvanometer circuit and G is the galvanometer resistance to be determined. (Numerical values are only examples).

| No. of obs | Currents | Resistance R in ohms | Shunt resistance S in ohms | Resistance $R_1$ in ohms | Deflections | $G=R_1$ Ohms | Mean G in Ohms |
|---|---|---|---|---|---|---|---|
| | Direct | 1000 | 0.1 | 0<br>80 | 10.6<br>5.3 | 80 | |
| 1 | Reverse | " | " | 0<br>81 | 10.4<br>5.2 | 81 | |
| | Direct | " | 0.14 | 0<br>80 | 10.2<br>5.1 | 80 | |
| 2 | Reverse | " | " | 0<br>79 | 10.4<br>5.2 | 79 | |
| | Direct | 750 | 0.16 | 0<br>" | " | " | |
| 3 | Reverse | " | " | 0<br>" | " | | |
| ect. | | | | | | | |